

# Note for section 4

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## Abstract

In this note, we discuss the localized Majorana mode in one dimension. We will first give some general remark on the condition for having localized Majorana modes and discuss a concrete field theory example. The field theory calculation follows the spirit of Jackiw and Rebbi zero mode.

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# 1 Introduction

Any complex fermion can be decomposed into two Majorana fermions

$$c = \frac{\chi_1 + i\chi_2}{2}, \quad c^\dagger = \frac{\chi_1 - i\chi_2}{2}. \quad (1)$$

In this note, we want to understand the general condition for having Majorana fermion modes and giving a concrete example that is able to realize it. Here, we consider a true fermionic system not fermions that are emergent in a bosonic system.

We notice that a Majorana fermion as an eigen mode of the Hamiltonian can appear *accidentally*. One extreme example is that when the Hamiltonian happens to be zero, all the eigen modes can be considered as Majorana fermions. Therefore it is worth to specify the meaning of “the existence of Majorana fermion” more carefully at the beginning to avoid ambiguity. In this note, “existence” automatically includes the robustness against *any symmetry allowed local perturbation* (it can be a sum of local terms). For the example above, those Majorana modes can be easily destroyed by arbitrarily small local chemical potentials and thus we will not refer to such kind of systems as supporting Majorana fermions. The same philosophy actually applies for all the discussion of topological phases.

Let us give some general remarks on Majorana fermion modes, their momentum and energy, some necessary conditions and the classification of their existence.

Suppose we have a system that can support Majorana modes, we want to find some constraints on their energy and momentum:

- Let the Majorana mode  $\chi$  carry an energy  $\varepsilon$ , which means

$$[H, \chi] = \varepsilon \chi \quad (2)$$

Note that the left-hand side is anti-Hermitian while the right-hand side is Hermitian. The only way for the equation to hold is to set  $\varepsilon = 0$ . Namely, Majorana modes can only carry zero energy.<sup>1</sup>

- If the system has a translation symmetry, the Majorana mode is allowed to carry a momentum  $k$ . Then we must have  $2k \bmod K = 0$ , where  $K$  is one of the reciprocal lattice vectors. This is because the Majorana mode is its own anti-particle, creating it twice must return us the original state.

This observation implies that any closed system with unbroken translation symmetry cannot support Majorana modes unless it is gapless.<sup>2</sup> To create Majorana modes in gapped systems, we have to consider either having topological defects in higher dimensions (e.g. vortex in 2+1-dimension or monopole in 3+1-dimension) or having an open boundary condition in one spatial

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<sup>1</sup>This situation can change when we consider Floquet systems  $U_F = \mathcal{T} e^{-i \int_0^T H(t) dt}$ , where the meaningful quantity is the quasi-energy  $U_F |\psi\rangle = e^{-iE} |\psi\rangle$ . If the system can support a Majorana mode  $\chi$  that carries a definite quasi-energy, it means

$$U_F \chi U_F^{-1} = e^{-i\varepsilon} \chi \quad (3)$$

Then, there are two ways for the equation to make sense,  $\varepsilon = 0, \pi$ .

<sup>2</sup>We need a physical assumption that the energy is a continuous function of the (many-body) momentum.

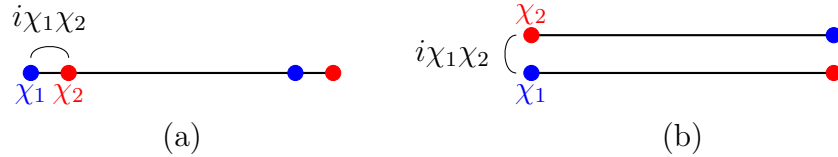


Figure 1: Two nearby localized Majorana fermions can cancel each other easily.

dimension.<sup>3</sup> And the Majorana mode, if exists, is expected to appear at the topological defects or the edge of the open chain, dubbed *localized* Majorana mode. If we aim at localized Majorana modes from the beginning, it is then natural to consider these setups (having topological defects or open boundary condition). The purpose of going through these arguments is to show that we have to consider these setups to find Majorana modes in gapped systems even though we start with complete generality.

Having a *gap* is also a necessary condition for the localized Majorana modes to become robust. This is because Majorana fermion has to appear in pairs to have a sensible Hilbert space. The existence of gap guarantees the coupling between them to vanish as  $e^{-L/\xi}$ , where  $L$  is their separation. There is another necessary condition, *superconductivity*, i.e. only fermion number mod 2 is conserved. If the system is not a superconductor and thus has an unbroken  $U(1)$  symmetry, then any eigenstate must have a definite particle number. No degeneracy implies no Majorana fermions. If there is a degeneracy between two eigenstates with the particle number  $n$  and  $n+1$  respectively, a chemical potential term, which is a local perturbation, can be added to lift the degeneracy and thus remove the Majorana fermions.

Provided these necessary conditions and no further symmetry constraint, the existence of localized Majorana modes is a  $\mathbb{Z}_2$  event. Let us illustrate it using one-dimensional systems, where the Majorana modes can only appear at the edges. If there are two Majorana fermions at each end of the open chain, a local coupling term can easily gap them out, as depicted in Fig. 1 (a). An equivalent viewpoint is to consider two parallel chains. If each has a single Majorana fermion, they can weakly interact and cancel each other, as depicted in Fig. 1 (b).

Consequently, each one-dimensional fermion Hamiltonian  $H$  can be characterized by a  $\mathbb{Z}_2$  quantity  $\mathcal{M}(H) = \pm 1$  with  $\mathcal{M}(H) = -1$  representing the existence of Majorana fermion. This is the *topological invariance* for the systems considered. If we use  $\oplus$  to stand for taking two chains together (with any allowed local perturbation between them), the above argument implies

$$\mathcal{M}(H_1 \oplus H_2) = \mathcal{M}(H_1)\mathcal{M}(H_2) \quad (4)$$

This definition of  $\mathcal{M}(H)$  is only for open chains. However, it can be also computed if we know data from closed chains, which is generally easier to handle. If we define  $P(H(L))$  as the ground state parity of  $H$  on a *closed* chain with length  $L$ , then  $\mathcal{M}(H)$  is related to  $P(H)$  as follows

$$P(H(L_1 + L_2)) = \mathcal{M}(H)P(H(L_1))P(H(L_2)). \quad (5)$$

This is an example of *bulk-edge correspondence*. The proof can be found in Appendix. A. The

<sup>3</sup>The reason that inserting topological defects, such as twisting the boundary condition at one link, does not work in one-dimension is that any such configuration is equivalent to a translationally-invariant one up to gauge transformation.

proof itself does not require the system to be non-interaction thus persists to hold even for interacting systems.

In the rest of this note, we will consider a concrete system, the Kitaev chain or one-dimensional p-wave superconductor. It has a trivial phase and a topological phase, where the localized Majorana fermion appears. We will understand it from different aspects.

## 2 The model

We consider spinless free fermion on a one-dimensional lattice

$$H = - \sum_j \left( w c_j^\dagger c_{j+1} + \Delta c_j c_{j+1} + h.c. \right) + \mu \sum_j (c_j^\dagger c_j - 1/2) \quad (6)$$

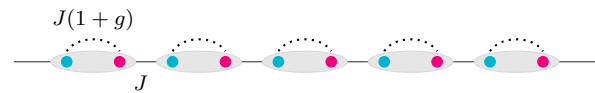
where  $w$  is the hopping,  $\Delta$  is the superconducting hopping and  $\mu$  is the on-site chemical potential. Without loss of generality, we assume  $\Delta > 0$ . It is convenient to rewrite the Hamiltonian in terms of Majorana fermions

$$c_j = \frac{\chi_{2j-1} + i\chi_{2j}}{2}, \quad c_j^\dagger = \frac{\chi_{2j-1} - i\chi_{2j}}{2}$$

and have

$$H = \frac{i}{2} \sum_j (w + \Delta) \chi_{2j} \chi_{2j+1} + (-w + \Delta) \chi_{2j-1} \chi_{2j+2} + \mu \chi_{2j-1} \chi_{2j}$$

For simplicity, we choose  $w = \Delta$  to have only nearest neighbor hopping term. And it is more convenient to adopt a different parametrization of the Hamiltonian

$$H = \frac{iJ}{2} \sum_j (1 + g) \chi_{2j-1} \chi_{2j} + \chi_{2j} \chi_{2j+1} \quad (7)$$


The system is gapless at  $g = 1$  and gapped elsewhere. The topological phase is expected to appear when the intra-cell coupling is weaker than the inter-cell coupling, i.e.  $g < 0$ .

## 3 Field theory analysis

$$\chi_{2j-1} = \frac{\psi_{L,j} + \psi_{R,j}}{\sqrt{2}}, \quad \chi_{2j} = \frac{\psi_{L,j} - \psi_{R,j}}{\sqrt{2}} \quad (8)$$

$$\{\psi_{a,j}, \psi_{b,k}\} = 2\delta_{ab}\delta_{jk} \quad (9)$$

$$\chi_{2j-1}\chi_{2j} = \psi_{R,j}\psi_{L,j}$$

$$\chi_{2j}\chi_{2j+1} = \frac{1}{2} (\psi_{L,j}\psi_{L,j+1} - \psi_{R,j}\psi_{R,j+1}) + \frac{1}{2} (\psi_{L,j}\psi_{R,j+1} - \psi_{R,j}\psi_{L,j+1})$$

## A Proof of Eq. (5)

We prove this relation by checking two cases  $\mathcal{M}(H) = \pm 1$  separately.

Let us first check the case with  $\mathcal{M}(H) = -1$ , namely the Hamiltonian on an open chain has a pair of Majorana fermions  $\eta, \tilde{\eta}$  localized at its two ends. Let  $|\psi_{\alpha=0,1}\rangle$  denote the two degenerate ground state and their the fermion parity is

$$P = s(L)(-i\eta\tilde{\eta}), \quad P|\psi_{\alpha}\rangle = s(L)(-1)^{\alpha}|\psi_{\alpha}\rangle, \quad (10)$$

where  $s(L)$  is the parity for the bulk part. When we close the chain by introducing a coupling between the two ends, the low energy effective Hamiltonian is

$$H = \frac{i}{2}u\tilde{\eta}\eta, \quad \text{Diagram: } \eta \bullet \xrightarrow{\text{dashed arrow}} \bullet \tilde{\eta} \quad (11)$$

which lifts the degeneracy between  $|\psi_{\alpha}\rangle$ . One can convenience himself that the fermion parity for the new ground state is

$$P(H(L)) = -s(L) \operatorname{sgn} u. \quad (12)$$

Now let us consider two such chains with length  $L_1$  and  $L_2$ . We can close each chain separately, the fermion parity of the ground state will be

$$P(H(L_1))P(H(L_2)) = s(L_1)s(L_2), \quad \text{Diagram: } \eta_1 \bullet \xrightarrow{\text{dashed arrow}} \bullet \tilde{\eta}_1 \quad \eta_2 \bullet \xrightarrow{\text{dashed arrow}} \bullet \tilde{\eta}_2 \quad (13)$$

We can also connect the two chains into a single one and then close it. The effective Hamiltonian is

$$H(L_1 + L_2) = \frac{i}{2}u(\tilde{\eta}_1\eta_2 + \tilde{\eta}_2\eta_1), \quad \text{Diagram: } \eta_1 \bullet \xrightarrow{\text{dashed arrow}} \bullet \tilde{\eta}_1 \quad \eta_2 \bullet \xrightarrow{\text{dashed arrow}} \bullet \tilde{\eta}_2 \quad (14)$$

The fermion parity of the ground state obtained in this way is

$$P(H(L_1 + L_2)) = s(L_1)(-i\eta_1\tilde{\eta}_1)s(L_2)(-i\eta_2\tilde{\eta}_2)$$

Notice that  $(-i\eta_1\tilde{\eta}_1)(-i\eta_2\tilde{\eta}_2) = -(-i\tilde{\eta}_1\eta_2)(-i\tilde{\eta}_2\eta_1)$ , we have

$$P(H(L_1 + L_2)) = -P(H(L_1))P(H(L_2)). \quad (15)$$

So the relation is true.

For the  $\mathcal{M}(H) = 1$  case, there is no Majorana fermion and we do not have worry about the subtleties mentioned above. It is not hard to see the relation also holds.

We would like to comment on the choice of  $u$  here.<sup>4</sup> During the proof, it seems that  $u$  can be chosen arbitrarily. However, it has to be consistent with the sign of couplings in the bulk to guarantee that the  $H$  defined on the closed chain satisfies periodic boundary condition, which is what we have assumed at the first beginning.

## References

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<sup>4</sup>We thank Nathanan Tantivasadakarn for helpful private discussion.